

Daily Quiz 8 (ELL782/ELL7282) - Computer Architectures

Instructions

- **Time:** 5 minutes
- Maximum marks: 10
- Number of questions: 10
- Each question carries **1 mark**.
- Questions marked **MCQ** have exactly one correct option.
- Questions marked **MSQ** may have one or more correct options. For an MSQ, select all correct options and no incorrect option.
- There is **no partial credit** for an MSQ.

Points: 10/10

Name * 

test

Entry Number * 

test

✓ **Correct** 1/1 Points

A full adder receives the inputs: $a = 1$, $b = 0$, $c_{in} = 1$ What is the output pair (sum, cout)? 

☐ (1, 1)

☐ (0, 0)

☒ (0, 1)☐ (1, 0)

✓ **Correct** 1/1 Points

Which statements are correct? 

Select all that apply.

☒ For a full adder: $\text{cout} = ab + a \cdot \text{cin} + b \cdot \text{cin}$

☒ The sum output of a full adder is 1 when an odd number of its three inputs are 1.

☒ A half adder adds two 1-bit inputs and produces a sum bit and a carry bit.

☐ The carry output of a full adder is 1 only when exactly two of its inputs are 1.

✓ **Correct** 1/1 Points

An n-bit ripple carry adder has a half-adder delay of 2 ns and a full-adder delay of 3 ns. Using:

Total delay = $t_h + (n - 1)t_f$

what is the delay of a 16-bit ripple carry adder? 

☐ 50 ns

☒ 47 ns

☐ 48 ns

☐ 45 ns

✓ **Correct** 1/1 Points

For $n = 256$, use the following simplified dominant time expressions:

- Ripple carry adder: n
- Optimally configured carry select adder: $2\sqrt{n}$
- Carry lookahead adder: $\log_2(n)$

Which tuple gives their respective time values? 

- ☐ (256, 64, 8)
- ☐ (256, 16, 8)
- ☐ (255, 32, 16)
- ☒ (256, 32, 8)

✓ **Correct** 1/1 Points

A carry select adder adds two 64-bit numbers using blocks of size k . Its total time is approximated by: $T(k) = k + 64/k$. Which block size gives the minimum time, and what is that time? 

- ☐ $k = 4, T = 20$
- ☐ $k = 32, T = 34$
- ☒ $k = 8, T = 16$
- ☐ $k = 16, T = 20$

✓ **Correct** 1/1 Points

For one bit position:

$$A_i = 0, B_i = 1, C_{in} = 1$$

Using:

$$g_i = A_i \cdot B_i$$

$$p_i = A_i \text{ XOR } B_i$$

$$C_{out} = g_i + p_i \cdot C_{in}$$

what is the tuple $(g_i, p_i, \text{sum}, C_{out})$? 

☐ (0, 1, 1, 0)

☐ (0, 0, 1, 1)

☒ (0, 1, 0, 1)

☐ (1, 0, 0, 1)

✓ **Correct** 1/1 Points

Which expression correctly gives the carry-out from the third bit position?



☐ $C_3 = g_3 + p_3g_2 + p_3p_2g_1 + p_1C_{in}$

☐ $C_3 = g_3 + p_3g_2 + p_2g_1 + p_3p_2p_1C_{in}$

☒ $C_3 = g_3 + p_3g_2 + p_3p_2g_1 + p_3p_2p_1C_{in}$

☐ $C_3 = g_3 + p_2g_2 + p_3p_2g_1 + p_3p_2p_1C_{in}$

✓ **Correct** 1/1 Points

Which statements correctly describe the operation of a carry select adder? 

Select all that apply.

☐ During the selection stage, the correct results for all blocks can always be selected simultaneously.

☒ Selection begins with the least significant block because its initial carry-in is known.

☒ The selected carry-out of one block determines which precomputed result is chosen for the next block.



Each block precomputes one result assuming carry-in 0 and another result assuming carry-in 1.

✓ **Correct** 1/1 Points

A lower block has: $GL = 1$, $PL = 0$. An upper block has: $GU = 0$, $PU = 1$. The combined block values are computed as:

$$G = GU + PU \cdot GL$$

$$P = PU \cdot PL \text{ and}$$

$$C_{out} = G + P \cdot C_{in}$$

What is the ordered pair: (C_{out} when $C_{in} = 0$, C_{out} when $C_{in} = 1$)? 

☐ (0, 1)

☐ (0, 0)

☐ (1, 0)

☒ (1, 1)

✓ **Correct** 1/1 Points

A carry lookahead computation starts with generate and propagate values for individual bits and combines equal-sized blocks in a tree: 1 bit $\rightarrow$ 2 bits $\rightarrow$ 4 bits $\rightarrow$ 8 bits $\rightarrow$ 16 bits $\rightarrow$ 32 bits. How many pairwise-combination stages are required beyond the initial per-bit level to obtain the generate and propagate values for the complete 32-bit block? 

☐ 32

☒ 5

☐ 4

☐ 6

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